



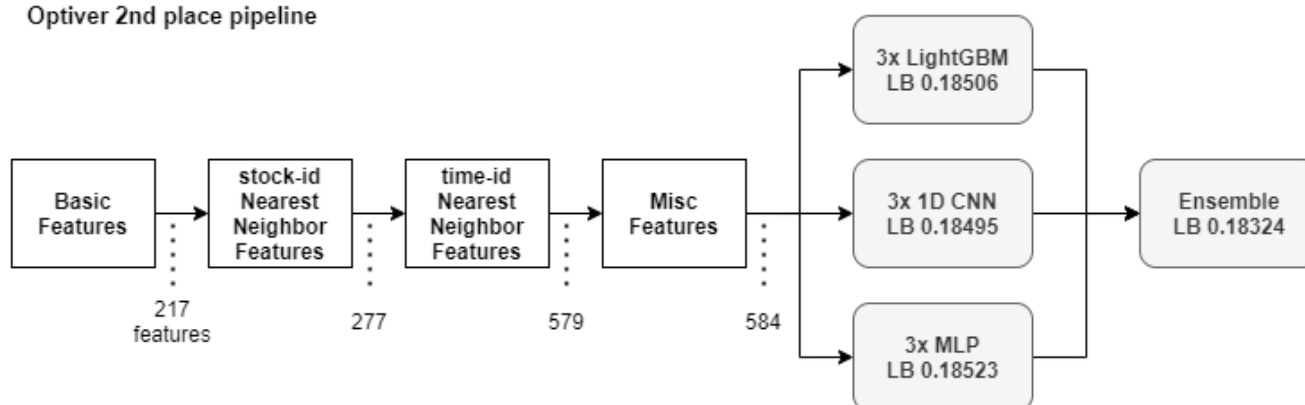
2022-01-11 Updated the title.

First of all, I would like to say a big thank you to Optiver and Kaggle for organizing a very interesting competition. I have never analyzed financial data before, but thanks to the very helpful tutorial notebook, I was able to fully immerse myself in the competition.

I don't know where I rank on the final leaderboard, but the points of my public 2nd place solution are as follows:

- Time-series cross-validation by reverse engineering of time-id order
- Nearest neighbor aggregation features (boost from 0.21 to 0.19)
- Blend of LightGBM, MLP, and MoA's 1D-CNN

Optiver 2nd place pipeline



You can also see my notebook here:

<https://www.kaggle.com/nyanpn/public-2nd-place-solution>

Now, I'd like to introduce the detailed solution.

Reverse engineering of time-id order

The prices in the competition data are normalized, but as someone pointed out in the discussion, you can use "tick size" to recover the real prices before normalization.

Furthermore, by compressing the time-id x stock-id price matrix to one dimension using t-SNE, I was able to recover the order of the time-id with sufficient accuracy.

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import glob
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